Meta-Reinforcement Learning for Uncertainty-Aware Integration of Scheduling and Control in Multi-Reactor Fermentation Processes

Project Proposal

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1 Introduction

The optimization and control of manufacturing systems represent a foundational challenge in chemical engineering. Traditional plant operations rely on an integrated scheduling and control (iSC) framework to connect production scheduling decisions with regulatory control loops. Traditional iSC architectures combining Model Predictive Control or feedback controllers with scheduling algorithms have proven highly effective in various applications; however, design choices can potentially lead to conservative operational boundaries [1]. Furthermore, their computational complexity scales exponentially with the number of variables in the system, making them computationally intractable in large-scale settings [2]. Reinforcement Learning (RL) overcomes these issues by developing an optimal policy through continuous interaction with the process environment during an offline training phase. Once deployed, it requires only a forward pass through a deep neural network to make a decision [3]. Despite the theoretical advantages of RL, translating these frameworks from isolated simulations to physical production plants has the inherant vulnerability of pervasive process uncertainty. A policy optimized exclusively on a nominal process model or a static dataset degrades drastically when exposed to real-world operational drift [4]. In chemical processing, this uncertainty typically occurs in three areas. First, raw material compositions, enzymatic activities, or substrate qualities vary significantly between batches. Second, plant model mismatch such as simplified lumped parameter models, Unmodeled latent dynamics and delays and estimated parameters. Finally, physical equipment degrades non-uniformly across a facility, leading to altered hydraulic profiles, localized heat transfer limitations, and shifting throughput constraints caused by catalyst poisoning, heat exchanger fouling, or mechanical equipment wear. Retraining a standard agent from scratch for every new operational regime or feedstock lot is prohibitively sample-inefficient and economically unviable for an actual physical plants. Conversely, standard robust control or broad domain randomization strategies often yield overly conservative policies, sacrificing substantial economic performance and throughput to guarantee baseline stability across the entire worst-case uncertainty horizon[5]. Meta-reinforcement learning (meta-RL) shifts the operational objective from optimizing a static control law to training the adaptive procedure itself. By exposing the agent to a structured family of simulated variations during an offline training phase, it develops a policy capable of rapidly inferring the latent characteristics of any specific process instance[6]. When deployed online, the agent requires only a minimal number of operational transitions to recognize current kinetic bottlenecks or feedstock compositions and adjust its integrated scheduling and control decisions accordingly[7].

2 Proposed Approach

2.1 POMDP Formulation and State Observability

We recast the fermentation environment as a Partially Observable Markov Decision Process (POMDP), defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{T}, \Omega, R, \gamma \rangle$, in order to explicitly separate the true reactor state from theinformation available to the agent. This separation is what motivates a meta-RL approach: the agent must learn to infer latent process conditions from a limited observation stream rather than assuming full state access.

The full state $s_t \in \mathcal{S}$ consists of the 15-dimensional concentration vector C_t (substrates, sugars, inhibitors, biomass, and products) together with the reactor mass M_t. Of these, only a subset is realistically measurable online and thus available to the agent as an observation $o_t \in \mathcal{O}$, drawn according to the observation function $\Omega(o_t \mid s_t, a_t)$:

- **Observable quantities:** soluble sugar concentrations (glucose C_G, xylose C_X), ethanol C_{Eth}, CO_2 evolution C_{CO_2}, reactor mass M, and the inflow/outflow rates $F_{\text{in}}, F_{\text{out}}$. These correspond to standard online instrumentation (HPLC or enzymatic assays, off-gas analysis, load cells, flow meters).
- **Partially observable quantities:** viable cell concentration C_{Cell} is typically inferred from optical density, which is noisy and cannot distinguish viable from non-viable biomass. We treat this as a delayed, high-variance observation rather than a clean state variable.
- **Latent (unobservable) quantities:** the solid-phase substrate pools (C_{CS}, C_{XS}, C_{LS}) and the associated enzyme adsorption states (C_{ef}, C_{eb}) are not measurable online in practice, requiring offline or destructive sampling. Similarly, the inhibitor pool (furfural C_F, HMF C_{HMF}, acetate C_{ACT}) is generally not available in real time and must be inferred indirectly from its downstream effect on reaction rates.

The *kinetic parameters* governing the rate expressions, including the pH-dependent modifiers η, Γ_G, and Γ_X, are also considered as a latent task context $z \sim p(z)$ sampled once per episode rather than as fixed constants. This variable is never directly observed by the agent and represents batch-to-batch biological variability (e.g., strain differences, feedstock composition, pH drift).

2.2 Meta-RL Architecture for iSC

Meta-RL addresses the POMDP above by decomposing the learning problem into two nested loops. In the *outer loop*, the agent is trained offline across a distribution of simulated fermentation tasks, each corresponding to a different sampled context $z \sim p(z)$ (i.e. a different feedstock composition, kinetic regime, or equipment condition). In the *inner loop*, for any single sampled task, the agent interacts with the environment over one or more episodes and must adapt its behavior using only the stream of observations, actions, and rewards it has collected so far — it never has direct access to z. Formally, this adaptation is expressed through a context-inference mapping $q_\psi(z_t \mid \tau_{1:t})$, which compresses the trajectory history $\tau_{1:t} = (o_1, a_1, r_1, \ldots, o_t)$ into a belief over the latent task variable, and a policy $\pi_\phi(a_t \mid o_t, z_t)$ conditioned on both the current observation and this inferred context. The outer-loop meta-training objective is to find encoder and policy parameters (ψ, ϕ) that maximize expected return after adaptation, averaged over the task distribution:

$$\max_{\psi,\phi} \; \mathbb{E}_{z\sim p(z)} \left[\mathbb{E}_{\tau\sim\pi_\phi(\cdot|q_\psi)} \left[\sum_t \gamma^t r_t \right] \right]. \tag{1}$$

Because the encoder is trained across many tasks rather than fit to a single process instance, it learns a general-purpose inference procedure: at deployment time, on a previously unseen batch or feedstock lot, the agent can update its belief over z_t within a handful of transitions and adjust its scheduling and control actions accordingly, without any gradient updates or retraining on the physical plant. The three methods below differ primarily in *how* q_ψ is represented and how the inner-loop adaptation is realized mechanically.

Candidate meta-RL methods. We consider three concrete instantiations of q_ψ and the outer training loop, each representing a distinct family of context-based meta-RL:

- **RL^2** RL^2 reformulates the meta-learning problem as a sequence-modeling task by embedding adaptation entirely within the recurrent forward pass of the policy[8]. The architecture utilizes a GRU whose hidden state is maintained across the entire task horizon without being reset between episodes. At each time step, the network receives the current observation along with the preceding action and reward. When trained end-to-end via an on-policy actor-critic algorithm like PPO, the policy gradient is computed over full multi-episode trajectories. This forces the recurrent weights to optimize the hidden state trajectory, effectively converting it into an implicit latent context representation (z_t) that encodes task-specific dynamics solely through historical dependencies.
- **PEARL** PEARL decouples task inference from control by utilizing an explicit, permutation-invariant context encoder[9]. This encoder processes a history of transitions from the current task to parameterize a Gaussian posterior distribution, $q_\psi(z \mid c_t)$, from which a distinct latent task context z is sampled. The actor and critic networks are then conditioned directly on this sample alongside the environmental observation. PEARL utilizes an off-policy framework based on Soft Actor-Critic (SAC), optimizing the encoder parameters by backpropagating the critic loss through the latent variables, while enforcing an information bottleneck via a Kullback-Leibler (KL) divergence penalty. This off-policy formulation enables high sample efficiency by reusing transitions from a global replay buffer across multiple gradient steps.
- **Gradient-based meta-RL** Unlike context-based approaches, model-agnostic meta-learning (MAML) eliminates the need for an auxiliary encoder or recurrent memory by adapting directly within the policy's parameter space[10]. The objective of the outer optimization loop is to discover a primary initialization of the policy weights (ϕ_0) that is highly sensitive to task-specific gradients. During adaptation to a new task, the policy undergoes one or a few standard policy-gradient updates using a small set of local rollouts, transforming the initialization into a task-adapted policy ($\phi_0 \to \phi_0'$). The meta-optimization step differentiates through this inner-loop gradient update, pushing the initial parameters toward a region where minimal data-driven adjustments yield maximum performance gains.

2.3 Uncertainty Modeling and Domain Randomization

The meta-policy is trained entirely offline against a distribution of simulated tasks $\mathcal{T} \sim p(\mathcal{T})$, where each task fixes a sampled realization of the latent parameters, $\boldsymbol{\theta} \sim p(\boldsymbol{\theta})$. The design of $p(\boldsymbol{\theta})$ is not arbitrary: it is anchored to the physically meaningful degrees of freedom of the biochemistry so that the induced task family spans the operating envelope a plant genuinely encounters, rather than perturbing parameters that have no clear physical referent.

What is randomized. We randomize four groups of quantities, each tied to a distinct physical source of plant-to-plant or batch-to-batch variation:

1. **Overall and fermentation-specific kinetic modifiers** η, γ_G, γ_X, which scale hydrolysis and sugar-

uptake rates and stand in for pH and enzyme-cocktail drift. In the simulator these enter as pre-computed Gaussian "bump" functions of pH—for example $\eta(\mathrm{pH}) = \eta_{\max} \exp\big(-(\mathrm{pH} - \mu_\eta)^2 / 2\sigma_\eta^2\big)$ with separate (μ, σ) pairs for the hydrolysis, glucose-uptake, and xylose-uptake channels—so that randomizing the operating pH (or directly randomizing η, γ_G, γ_X around their nominal values) induces a physically coherent, correlated shift across all three rate channels simultaneously, rather than perturbing them independently.

2. **Toxin sensitivity coefficients** $K_{i,t}$ governing furfural, acetic-acid, and HMF inhibition of glucose and xylose uptake, capturing strain-dependent tolerance. These are randomized as multiplicative factors on the half-saturation/inhibition constants in the Monod-type inhibition terms, so that a single draw can represent, e.g., a more furfural-tolerant yeast strain without decoupling the (biologically linked) glucose- and xylose-channel tolerances.
3. **Enzyme adsorption capacities and affinities** $E_{\max,j}, K_j$ for the three enzyme components (cellulase-on-cellulose, cellulase-on-lignin/nonproductive binding, xylanase-on-xylan), representing lot-to-lot enzyme efficacy. These parameterize the Langmuir-type adsorption equilibrium solved per reactor at every rate evaluation; randomizing them changes how much of the dosed enzyme is bound and catalytically active for a given solids loading.
4. **Feedstock composition**, i.e. the inlet solid fractions and cellulose/xylan/lignin split together with the glucose/xylose ratio in $\mathbf{C}_{\mathrm{in}}$, representing batch-to-batch lot variability in the raw biomass feedstock.

For each meta-training task the following guidelines are followed during randomization: (i) draw each group's parameters from independent bounded priors whose ranges are based on heuristics, rather than allowing unbounded excursions that would force the policy toward unnecessary conservatism; (ii) impose correlated draws within a group where the underlying mechanism co-varies such as enzyme loading and adsorption capacity are drawn jointly rather than independently, since in the physical system a single upstream cause (pH shift, enzyme lot) manifests as simultaneous, coupled changes across several of these coefficients; (iii) $\boldsymbol{\theta}$ remained as the same value for the entire duration of one task episode so that the within-episode transition history c_t is informative about a single underlying regime; and (iv) make sure the evaluation contains enough OOD cases to avoid overfitting.

3 Evaluation Plan

3.1 Baseline Comparisons

To directly isolate the benefits of meta-learning, the evaluation focuses on a comparison between the proposed adaptive framework and a non-adaptive baseline, keeping the underlying safety constraints identical across both. Specifically, the proposed meta-policy is benchmarked against a non-adaptive Safe Deep RL agent. This baseline incorporates the exact same safety architecture such as reward penalties and safety filters, but is trained strictly on the nominal parameterization $\boldsymbol{\theta}_0$ and deployed without any context inference or memory mechanisms.

3.2 Uncertainty Stress-Testing

To test the core claim of rapid online adaptation, three mid-episode disturbance scenarios were introduced to force the agent to recognize and respond to sudden regime changes. The first is an abrupt step change in inlet composition. The second is a sudden degradation of enzyme efficacy, which restricts adsorption capacities and stalls downstream fermentation. The third is a gradual reactor mass drift that slowly alters the residence-time structure and mass balances. For each scenario, we evaluate adaptation speed and resilience

of the agent by measuring final reward value, the number of time required to recover performance and constraint violations.

References

[1] Logan D. R. Beal, Damon Petersen, David Grimsman, Sean Warnick, and John D. Hedengren. Integrated scheduling and control in discrete-time with dynamic parameters and constraints. *Computers & Chemical Engineering*, 115:361–376, 2018. doi: 10.1016/j.compchemeng.2018.04.010.

[2] Michael Baldea, Juan Du, Jungup Park, and Iiro Harjunkoski. Integrated production scheduling and model predictive control of continuous processes. 61(12):4179–4190. doi: 10.1002/aic.14951.

[3] Daniel Rangel-Martinez and Luis Ricardez-Sandoval. Hybrid deep reinforcement learning agent for online scheduling and control for chemical batch plants. *IFAC-PapersOnLine*, 59(6):259–264, 2025. ISSN 2405-8963. doi: https://doi.org/10.1016/j.ifacol.2025.07.155. 14th IFAC Symposium on Dynamics and Control of Process Systems, including Biosystems DYCOPS 2025.

[4] T. Savage, D. Zhang, M. Mowbray, and E. A. del Río Chanona. Model-free safe reinforcement learning for chemical processes using gaussian processes. *IFAC-PapersOnLine*, 54(3):504–509, 2021. ISSN 2405-8963. doi: https://doi.org/10.1016/j.ifacol.2021.08.292. 16th IFAC Symposium on Advanced Control of Chemical Processes ADCHEM 2021.

[5] G. Tiboni, P. Klink, J. Peters, T. Tommasi, C. D'Eramo, and G. Chalvatzaki. Domain randomization via entropy maximization. *arXiv preprint arXiv:2311.01885*, 2023. doi: https://doi.org/10.48550/arxiv.2311.01885.

[6] S. Sæmundsson, K. Hofmann, and M. P. Deisenroth. Meta reinforcement learning with latent variable gaussian processes. In *Proceedings of the Thirty-Fourth Conference on Uncertainty in Artificial Intelligence*, 2018. doi: https://doi.org/10.48550/arxiv.1803.07551.

[7] H. Fu, H. Tang, J. Hao, C. Chen, X. Feng, D. Li, and W. Liu. Towards effective context for meta-reinforcement learning: an approach based on contrastive learning. *arXiv preprint arXiv:2009.13891*, 2020. doi: https://doi.org/10.48550/arxiv.2009.13891.

[8] Y. Duan, J. Schulman, X. Chen, P. L. Bartlett, I. Sutskever, and P. Abbeel. Rl^2: Fast reinforcement learning via slow reinforcement learning. In *Proceedings of the 33rd International Conference on Machine Learning*, 2016. doi: https://doi.org/10.48550/arxiv.1611.02779.

[9] K. Rakelly, A. Zhou Resolving, C. Finn, S. Levine, and D. Quillen. Efficient off-policy meta-reinforcement learning via probabilistic context variables. In *Proceedings of the 36th International Conference on Machine Learning*, 2019. doi: https://doi.org/10.48550/arxiv.1903.08254.

[10] C. Finn, P. Abbeel, and S. Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *Proceedings of the 34th International Conference on Machine Learning*, 2017. doi: https://doi.org/10.48550/arxiv.1703.03400.